

FRENCH CHEESE SHOP

AI in Requirements Engineering

A French cheese shop demo surfaces three AI RE challenges: interpreting vague requests, using cheese knowledge and shop context, and evaluating behavior under ambiguity.

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**Ambiguity
Data
Evaluation**



Photo: jackmac34 / Pixabay

SESSION FOCUS

Learning Outcomes

1

Interpret vague requests

Vague requests force the system to infer missing preferences and constraints.

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3

Evaluate ambiguity

Judge how the system handles ambiguity, not only whether the answer is correct.

CUSTOMER QUERY

A Vague Customer Request



“I want something
like Brie,
but stronger.”

Easy for a shopkeeper who knows the cheeses
and the customer. Hard for a system without both.

Photo: lee_2 / Pixabay

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Good answer

Best match, shortlist,
or clear explanation

CONCEPTUAL SHIFT

Traditional RE vs AI-Supported RE

The key shift is from specifying functions to defining behavior under uncertainty.

Traditional

- Search cheeses by type
- Validate fields and stock
- Same input, same output
- Success = feature works

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Requirements go beyond functions to specify how the system interprets vague requests.

Live Demo

The demo surfaces three issues: vague requests, real shop context, and uncertainty handling.

Open the demo

french-cheese-shop-
demo.survivejs.workers.dev

Scan the QR code



For AI systems, requirements should specify behavior under uncertainty.

Questions and Comments

References

Cited sources

[1]

B. Nuseibeh and S. Easterbrook. "Requirements Engineering: A Roadmap." In The Future of Software Engineering, ICSE 2000.

[2]

N. Kiyavitskaya, N. Zeni, L. Mich, and D. M. Berry. "Requirements for tools for ambiguity identification and measurement in natural language requirements specifications." Requirements Engineering, 13(3), 2008.

[3]

K. Ahmad, M. Abdelrazek, C. Arora, M. Bano, and J. Grundy. "Requirements engineering for artificial intelligence systems: A systematic mapping study." Information and Software Technology, 158, 2023.